GGG-W-290C
April 5, 1985
SUPERSEDING
GGG-W-290B
September 16, 1971

FEDERAL SPECIFICATION

WHEELS, ABRASIVE, SNAGGING

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers coarse abrasive snagging wheels, cones, and plugs for rapid stock removal without regard for close tolerances or surface finish requirements.

1.2 Classification. The snagging wheels covered by this specification shall be of the following types, classes, bonds, and sizes, as specified (see 6.2.1 and table I).

Type 1 - Wheel, straight (fig. 1).
Type 6 - Wheel, straight cup (fig. 2).
Type 11 - Wheel, flaring cup (fig. 3).
Type 16 - Cone, curved side (fig. 4).
Type 17 - Cone, straight side, square tip (fig. 5).
Type 17R - Cone, straight side, round tip (fig. 6).
Type 18 - Plug, square end (fig. 7).
Type 18R - Plug, round end (fig. 8).
Type 19 - Plug, conical end, square tip (fig. 9).
Type 19R - Plug, conical end, round tip (fig. 10).
Type 27 - Wheel, depressed center, flat rim (fig. 11).
Type 28 - Wheel, depressed center, saucer shaped rim (see fig. 12).

Class A - Aluminum oxide abrasives for snagging wheels (see table I).
Class C - Silicon carbide abrasives for snagging wheels (see table I).
Class Z - Zirconia abrasives for snagging wheels (see table I).

Bond B - Resinoid (see table I).
Bond V - Vitrified (see table I).

Size - The size snagging wheel to be furnished for a specific procurement shall be in accordance with table I and figures 1 through 12 as applicable to the type wheel required.

FSC 3460, 5130 and 5345

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TABLE I. Wheel selection guide - type, size, class, and bond

Wheel Types		Size, inches (overall wheel diameter)																						
		1	1-1/4	1-1/2	1-3/4	2	2-1/4	2-1/2	2-3/4	3	4	5	6	7	8	9	10	12	14	16	18	20	22	24	30
1	class				A,Z					A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,Z	A,C,Z	A,Z	A,C,Z	A,C,Z	C,Z	C,Z	A,C,Z	A,C,Z
	bond				B					B	B,V	B	B,V	B	B	B	V	B,V	B,V	B,V	B,V	B,V	V	B,V	B
6	class										A,C,Z	A,C,Z	A,C,Z												
	bond										B	B	B												
11	class									A	A,C,Z	A,C,Z	A,C,Z												
	bond									B	B	B	B												
16	class	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z															
	bond	B,V	B,V	B,V	B	B	B	B,V	B	B															
17	class	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z															
	bond	B	B	B,V	B	B	V	V	B	B															
17R	class	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z															
	bond	B	B	B	B	B	B	B	B	B															
18	class	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z															
	bond	B	B	B,V	B	B,V	B	B	B	V															
18R	class	A,C	A,C	A,C	A,C	A,C	A,C	A,C	A,C	A,C															
	bond	B,V	B	B	B	B,V	B	B	B	V															
19	class	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z															
	bond	B	B,V	B	B	B,V	B	B	B	V															
19R	class	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z	A,C,Z															
	bond	B	B	B	B	B	B	B	B	B															
27	class									A,Z	A,Z			A,Z		A,Z									
	bond									B	B			B		B									
28	class													A,C,Z		A									
	bond													B		B									

Notes: Letters located at the intersection of the type and size columns indicate the abrasive class and wheel bond available for the wheel size and types specified. The class and bond are described as follows:

Class - A - Aluminum oxide
C - Silicon carbide
Z - Zirconia

Bond - B - Resinoid
V - Vitrified

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2. APPLICABLE DOCUMENTS

2.1 Government Publications. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standards

FED-STD-H28 - Screw Thread Standards for Federal Services.
FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification

MIL-A-3816 - Abrasives and Abrasive Products, Packaging of.

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Federal Regulation

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

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(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI) Standards

ANSI B7.1 - Safety Code for the Use, Care, and Protection of Abrasive Wheels.

ANSI B74.2 - Specification for Shapes and Sizes of Grinding Wheels and for Shapes, Sizes, and Identification of Mounted Wheels.

ANSI B74.12 - Size of Abrasive Grain Grinding Wheels, Polishing and General Industrial Uses, Specifications for the.

ANSI B74.13 - Marking for Identifying Grinding Wheels and Other Bonded Abrasives.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

American Society For Testing and Materials (ASTM)

ASTM D 3951 - Commercial Packaging, Standard Practice for.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification will take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the supplier shall furnish two wheels of each type being procured for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification.

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3.2 Design. The snagging wheels shall be new and one of the manufacturer's current models that will meet the requirements of this specification. Unless otherwise specified (see 6.2.1), types (shapes) shall conform to ANSI B74.2. (see figures 1 through 12).

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the snagging wheels. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2 ¹). In this specification, all measurements, dimensions, sizes and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.2 Safety and health requirements. All snagging wheels covered by this specification shall conform to the applicable requirements of ANSI B7.1 and OSHA 2206.

3.3 Construction. All abrasive snagging wheels shall be constructed in accordance with established standards to provide for safe utilization during snagging operations. The wheel shall be complete so that when mounted on the grinding machine it can be used for the operation for which it is being purchased. The wheels shall be devoid of any defects or characteristics that will preclude conformance with any of the requirements under paragraph 3, and inspections and tests under paragraph 4.

3.4 Materials.

3.4.1 Reclaimed materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable.

3.4.2 Aluminum oxide abrasives. Snagging wheels made of aluminum oxide abrasives are recommended for grinding high tensile strength ferrous metals such as carbon steel, alloy steel, wrought iron, and annealed malleable iron. Aluminum oxide shall be represented by the standard letter "A" in accordance with ANSI B74.13.

3.4.3 Silicon carbide abrasives. Snagging wheels constructed of silicon carbide are intended for use in grinding low tensile strength materials such as gray iron, brass, and bronze. Silicon carbide shall be represented by the standard letter "C" in accordance with ANSI B74.13.

3.4.4 Zirconia abrasives. Snagging wheels constructed of zirconia abrasives are used to grind medium to high tensile strength ferrous materials. Zirconia shall be represented by the standard letter "Z".

3.4.5 Abrasive (grain) size. The nominal abrasive grain size shall be controlled within the limits for sizing of abrasives in accordance with ANSI B74.12. The abrasive grain size to be furnished shall be represented by the standard numbers 14, 16, 20, 24 and 30 in accordance with ANSI B74.3.

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Unless otherwise specified, the abrasive grain size to be furnished shall be 20. When specified (see 6.2.1), abrasive grain size shall be 14, 16, 24 or 30.

3.4.6 Grade. The grade (hardness) of the snagging wheels to be furnished shall be represented by the standard letters L, M, N, O, P, Q, R, S, T, and U in accordance with ANSI B74.13. Unless otherwise specified, the grade of the snagging wheels to be furnished shall be O. When specified (see 6.2.1), the grade shall be L, M, N, P, Q, R, S, T, or U.

3.4.7 Structure. The structure (porosity) of the snagging wheels shall be represented by the standard numbers 4,5, 6, 7, 8, and 9 in accordance with ANSI B74.13. Unless otherwise specified, the structure of the snagging wheels to be furnished shall be 6. When specified (see 6.2.1), the structure shall be 4, 5, 7, 8 or 9 as specified.

3.4.8 Bond. The bond (binder) of the snagging wheels shall be either resinoid or vitrified and shall be represented by the standard letters "B" and "V" respectively in accordance with ANSI B74.13.

3.5 Bushings. Bushings made of lead, fiber, plastic, or other suitable material shall be installed in the arbor hole of each type 1 abrasive wheel.

3.6 Threaded bushings. A threaded bushing shall be securely anchored into the arbor hole of each type 6, 11, 16, 17, 17R, 18, 18R, 19, and 19R wheel, cone, and plug. The direction of the threads shall be such that to remove the wheel from a grinder spindle the wheel must be turned in the same direction the spindle rotates when in use. The threads shall be either unified coarse thread (UNC) series or unified fine thread (UNF) series and shall have a class 2B fit in accordance with FED-STD-H28. Unless otherwise specified, the hole diameter and threads of the wheel, cone, or plug to be furnished shall be 5/8 - 11 UNC. When specified (see 6.2.1), the diameter and threads shall be either 1/2 - 13 UNC or 3/8 - 24 UNF. Each bushing shall be counterbored or countersunk to a depth of one thread.

3.7 Adaptors or flanges. Adaptors or flanges shall be provided with each type 27 and 28 depressed center wheel which does not have a mount permanently attached. The adaptors and flanges shall meet the design and dimensions specified in ANSI B7.1.

3.8 Blotters. Sufficient cushions of blotting paper, compression washers, or other suitable material of suitable size and thickness shall be furnished for each wheel, cone, or plug as specified in accordance with ANSI B7.1.

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3.9 Operating speed. Unless otherwise specified, the snagging wheels shall be designed to provide safety of operation at any speed up to 6500 surface feet per minute (SFPM) for vitrified bonded wheels and speeds up to 9500 SFPM for resinoid bonded wheels. When specified (see 6.2.1), the snagging wheels shall be designed for higher operating speeds as required.

3.10 Reinforcement. All type 27 and type 28 depressed center snagging wheels shall be reinforced with fabric, filament, or other suitable material. Unless otherwise specified, resinoid bonded wheels designed to operate at 12,500 SFPM or higher shall be reinforced. When specified (see 6.2.1), resinoid bonded wheels designed to operate below 12,500 SFPM shall be reinforced.

3.11 Trueness. Each snagging wheel shall be trued to size on all surfaces after firing or curing to ensure the wheel running true with the machine spindle.

3.12 Balance. The wheels shall be furnished in static balance in accordance with standard industry practice. Wheels which do not have inherent balance but are within the limits of counterbalancing based upon the size, structure, and type of application of the wheel, may be balanced to the limits recognized by industry by introducing fine abrasive powder or other suitable material into the wheel in such a manner as to not affect its serviceability. Balancing methods which may affect wheel strength and safety will be considered cause for rejection. Lead shall not be used to counterbalance wheels.

3.13 Marking. Each wheel shall be marked or labeled in a legible manner with the following data:

(a) Size (diameter, thickness, and hole diameter or thread).
(b) Maximum recommended operating speed.
(c) Manufacturer's name or trademark.
(d) Identification data (sequence of marking) as required by the standard marking system of ANSI B74.13.

3.14 Identification marking. When specified (see 6.2.1), the snagging wheel shall be marked or labeled with the appropriate National Stock Number furnished by the preparing activity.

3.15 Performance characteristics. The wheels shall be capable of removing large amounts of material and shall withstand the rotational stresses encountered at the maximum recommended operating speed in surface feet per minute, indicated by the manufacturer, as applicable to the type and size of abrasive wheel.

3.16 Workmanship. The wheels, cones, and plugs shall be uniform in quality and free from any imperfections which would cause breaking, cracking, or chipping under operating conditions.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).

b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be applied to the preproduction wheels or initial production wheels noted in 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.3 and the test in 4.4, as applicable. Failure of the item to pass the first article examination and test shall be cause for rejection.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to production wheels offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) and (b) as follows:

(a) Examination (see 4.3).

(b) Preparation for delivery inspection (see 4.5).

4.3 Examination. Each wheel shall be visually examined, inspected, and measured to ensure compliance with the requirements herein.

4.4 Test. The snagging wheels shall be tested in accordance with the requirements of ANSI B7.1.

4.5 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), packaging, packing, and marking shall be accomplished in accordance with MIL-A-3816.

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6. NOTES

6.1 Intended use. Type 1 resinoid and type 1 vitrified bonded wheels are intended for use on floor stand, bench stand, pedestal, or swing frame grinders. Type 1, 6, 11, 16, 17, 17R, 18, 18R, 19, 19R, 27, and 28 resinoid and vitrified bonded snagging wheels, cones, and plugs are intended for use on portable grinders.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, bond, and size (diameter, thickness, size arbor hole) wheel required (see 1.2).

c. First article, if required (see 3.1).

d. Type (shape), if different (see 3.2).

e. If abrasive wheel is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

f. Abrasive (grain) size required, if different (see 3.4.5).

g. Wheel grade required, if different (see 3.4.6).

h. Wheel structure required, if different (see 3.4.7).

i. Diameter of arbor hole and thread, if different (see 3.6).

j. Operating speed required, if different (see 3.9).

k. Reinforced wheel to operate below 12,500 SFPM, if required (see 3.10).

l. Marked with National Stock Number, if required (see 3.14).

m. First article inspection, if different (see 4.2.1).

n. Quality conformance inspection, if different (see 4.2.2).

o. Level of packaging, packing, and marking, if different (see 5.1).

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6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Cross-reference. Cross reference of classification changes between this specification and the preceding issue are as follows:

Deleted	Added
Type 4	Type 27
Type 5	Type 28

6.5 Standards shapes of grinding wheels. Figures 1 through 12 depict the standard shapes of grinding wheel shapes and faces which commonly apply to the grinding wheel types.

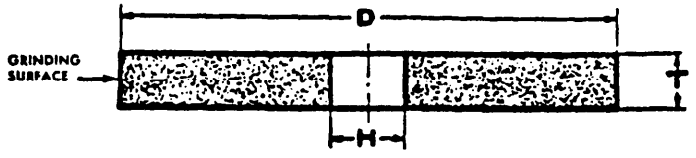


Figure 1. Type 1 - Wheel, straight.

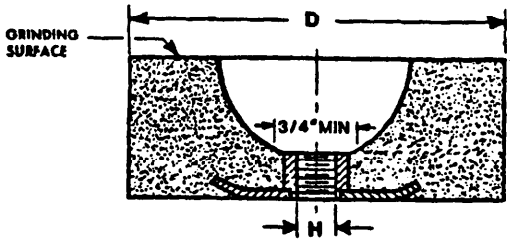


Figure 2. Type 6 - Wheel, straight cup.

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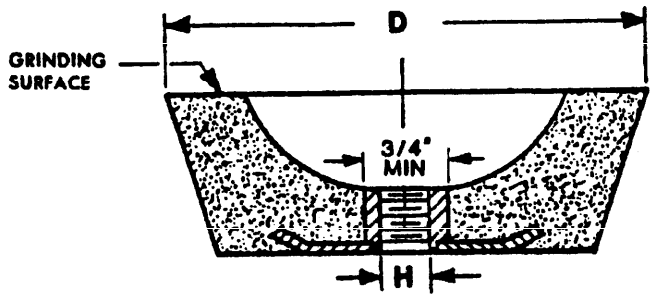


Figure 3. Type 11 - Wheel, flaring cup.

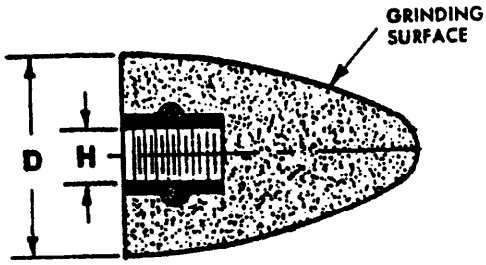


Figure 4. Type 16 - Cone, curved side.

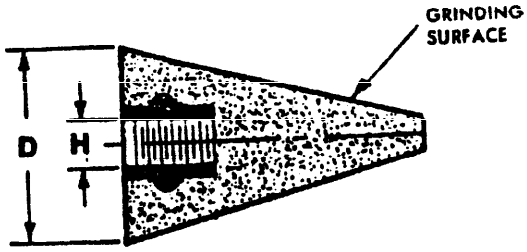


Figure 5. Type 17 - Cone, straight side, square tip.

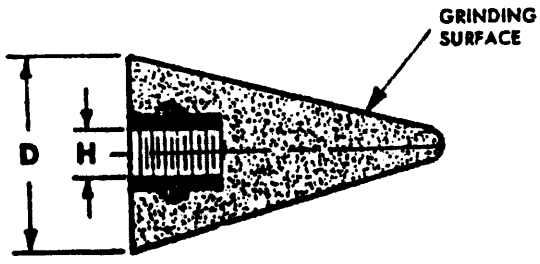


Figure 6. Type 17R - Cone, straight side, round tip.

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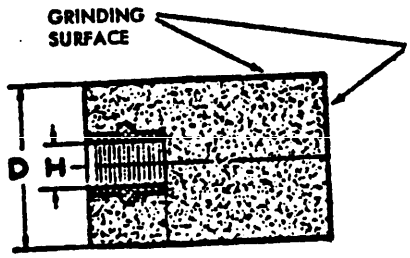


Figure 7. Type 18 - Plug, square end.

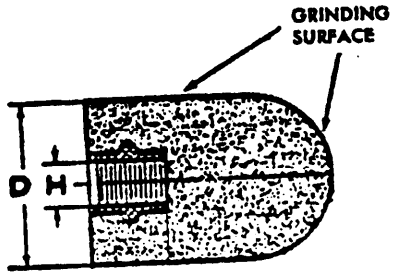


Figure 8. Type 18R - Plug, round end.

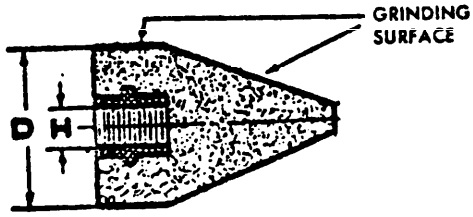


Figure 9. Type 19 - Plug, conical end, square tip.

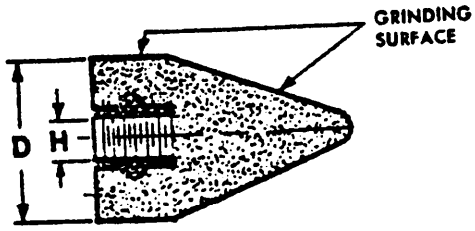


Figure 10. Type 19R - Plug, conical end, round tip.

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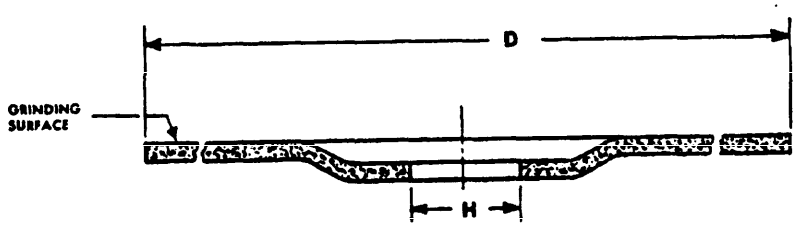


Figure 11. Type 27 - Wheel, depressed center, flat rim.

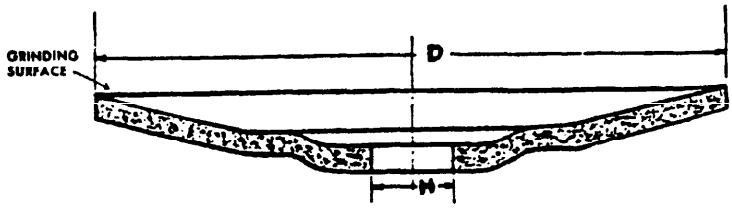


Figure 12. Type 28 - Wheel, depressed center, saucer shaped rim.

MILITARY INTERESTS:

Custodians

Army - AL
Navy - SH
Air Force - 99

Review Activities

Army - GL
Air Force - 84
DLA - GS, DH

User Activities

Army - ME
Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

DLA - IP

DoD Project 3460-0104

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
GGG-W-290C	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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